

Copy or Collaborate in a Changing Community

Team 11

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Our Inspiration

<https://doi.org/10.1038/s44260-025-00058-8>

Copy or collaborate? How networks impact collective problem solving

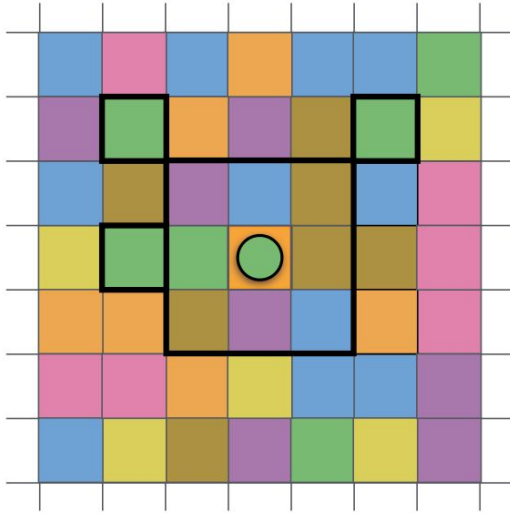


Check for updates

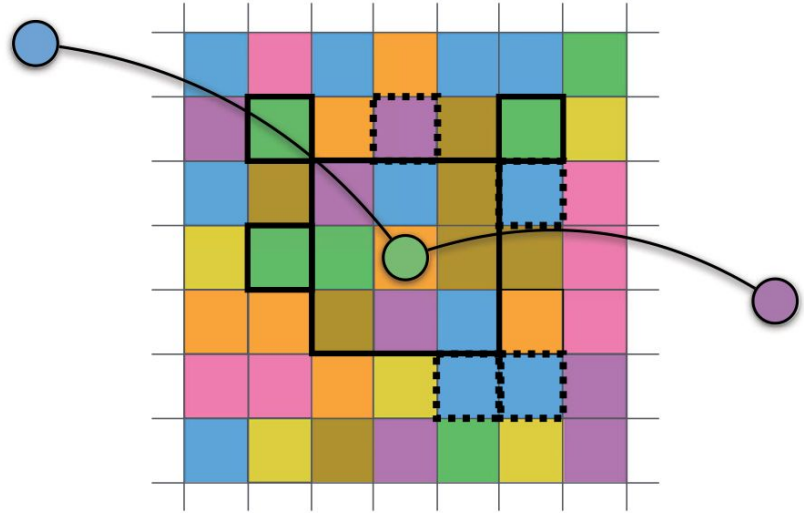
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Agents on a Grid

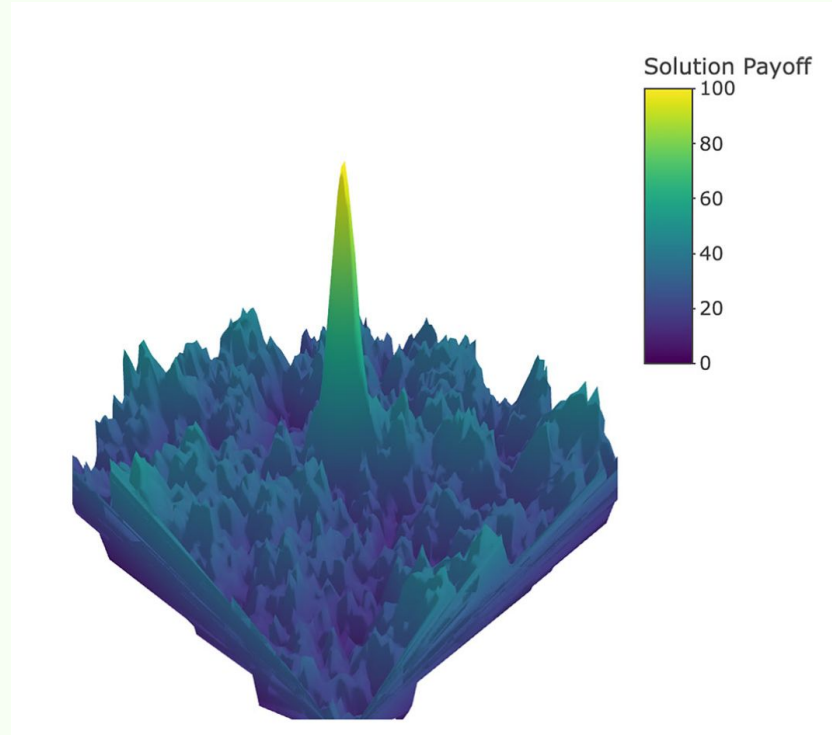
A.



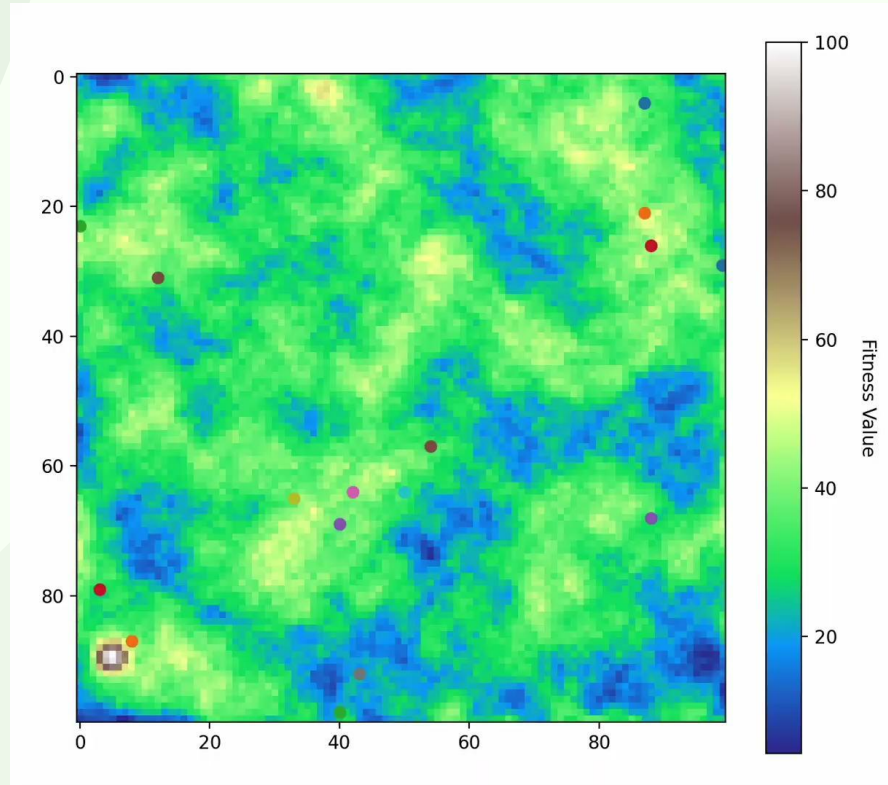
B.



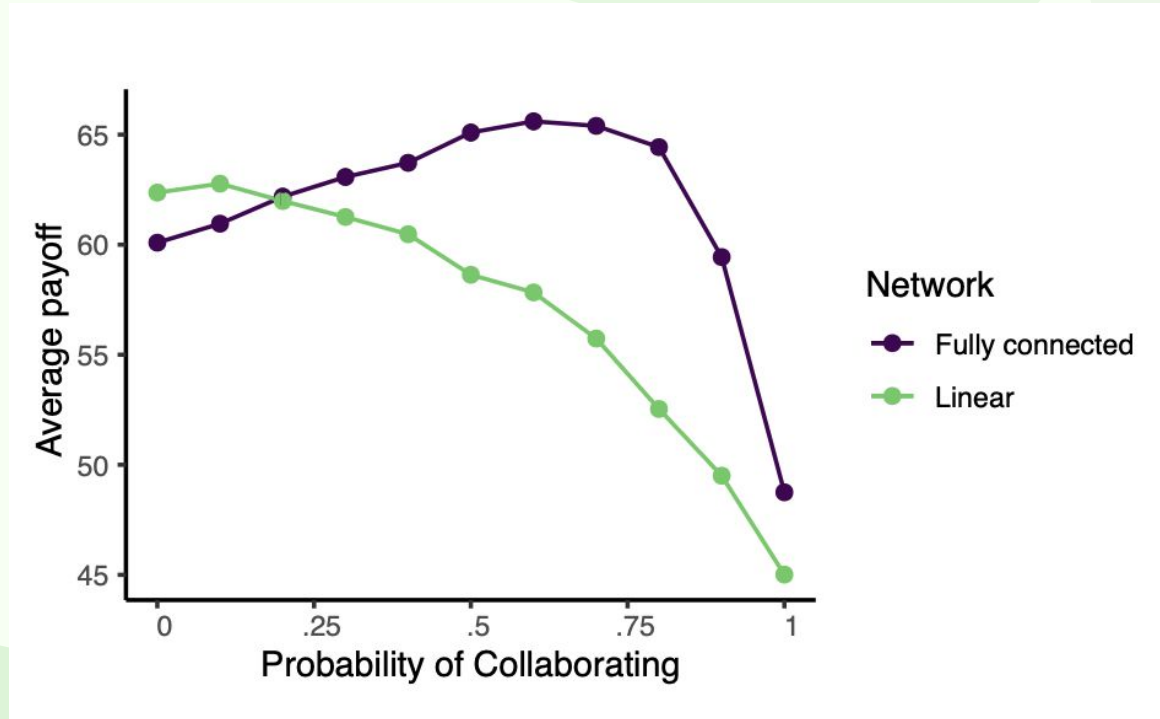
Fitness Landscape



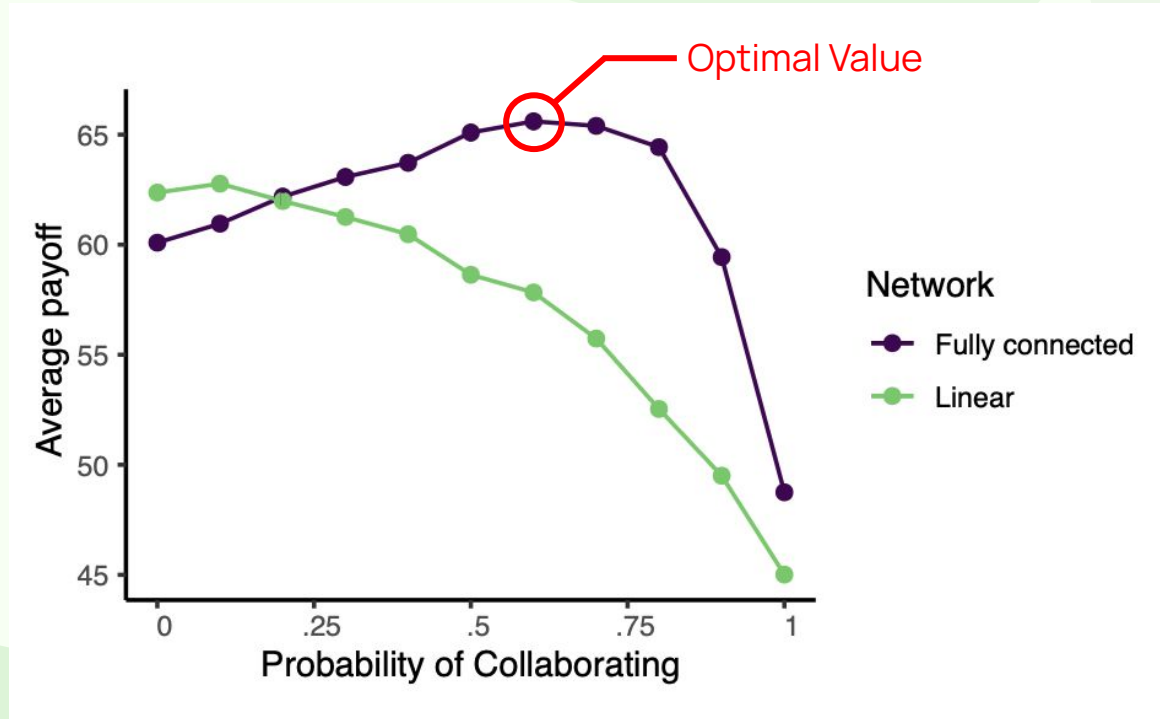
Movement each Step



Results of Paper



Results of Paper



What happens if
we add turnover?

Research Question

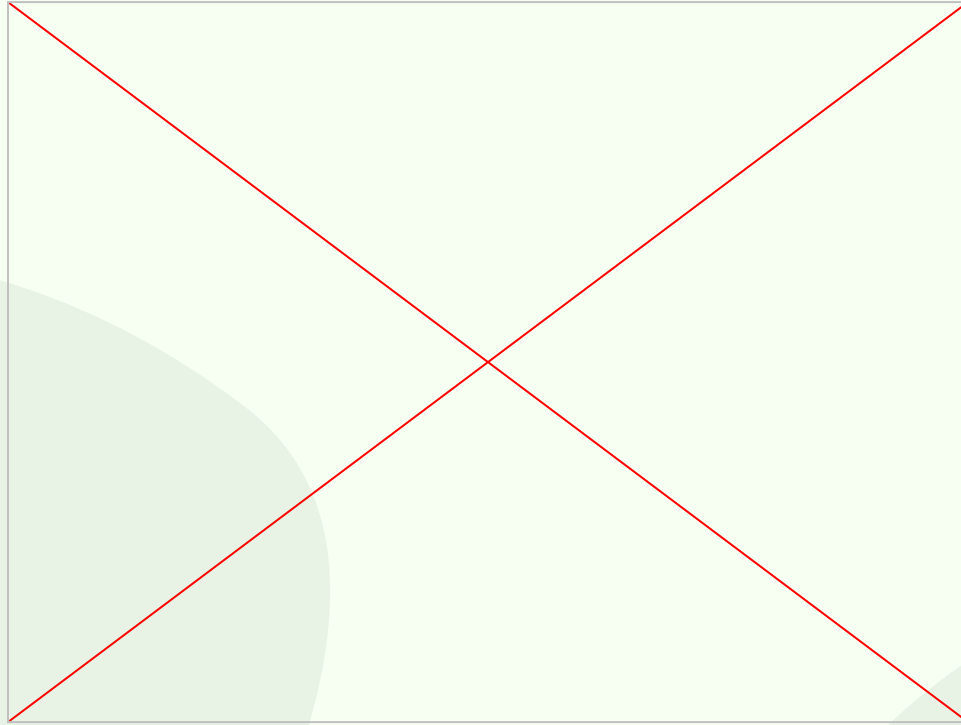
How does a dynamic agent network involving replacement of agents affect the optimal collaboration-copying ratio between agents?

Hypotheses

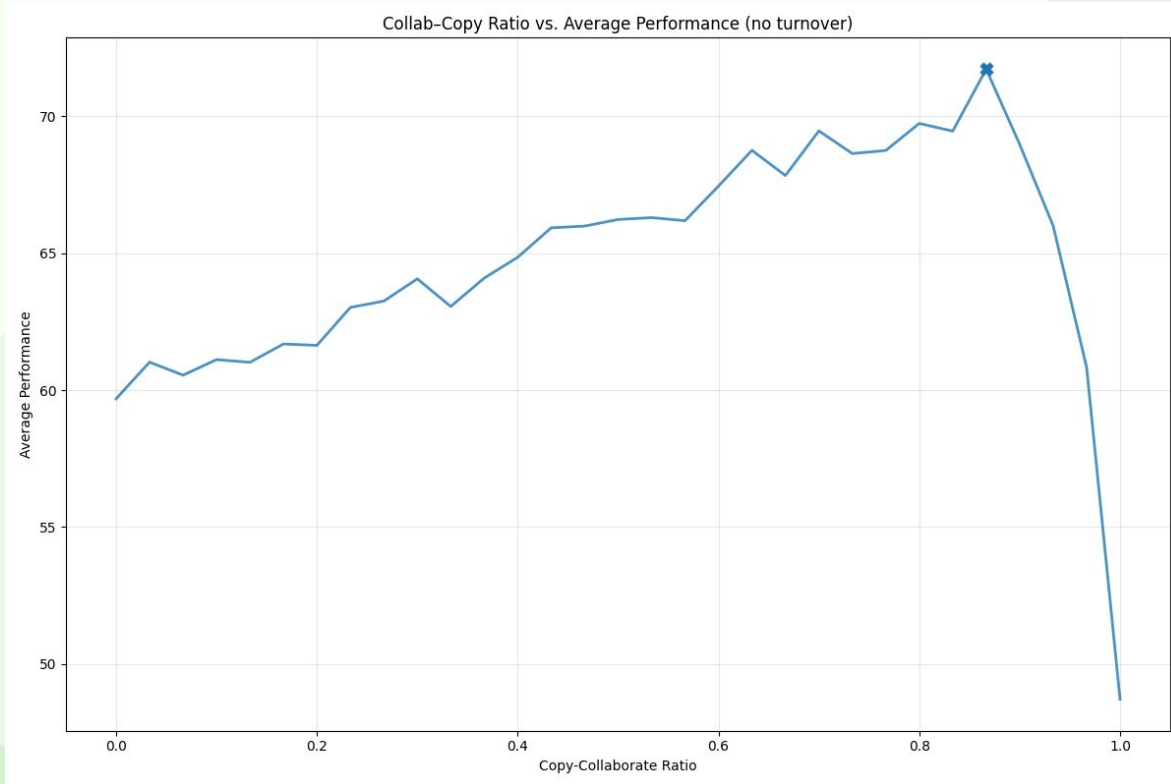
H1: Adding **turnover** has a statistically significant effect on the **optimal collaboration-copying ratio**.

H0: Adding **turnover** has no statistically significant effect on the **optimal collaboration-copying ratio**.

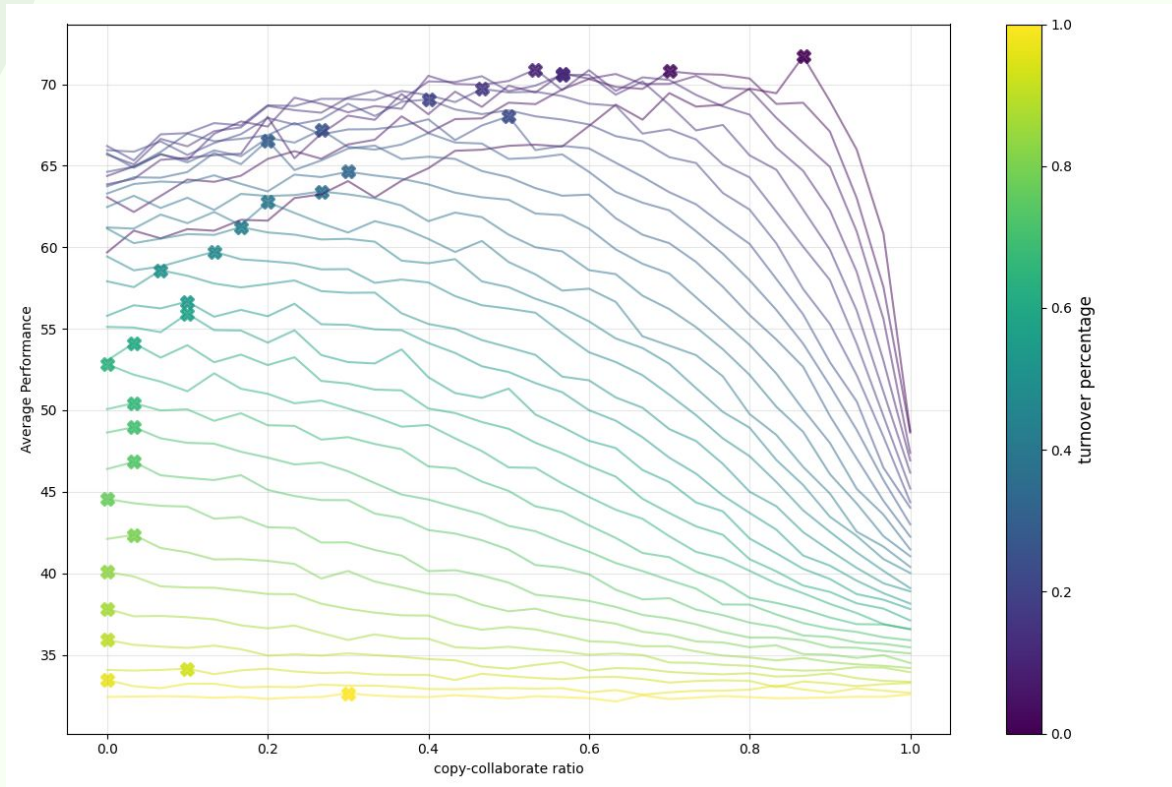
Movement Each Step with Turnover



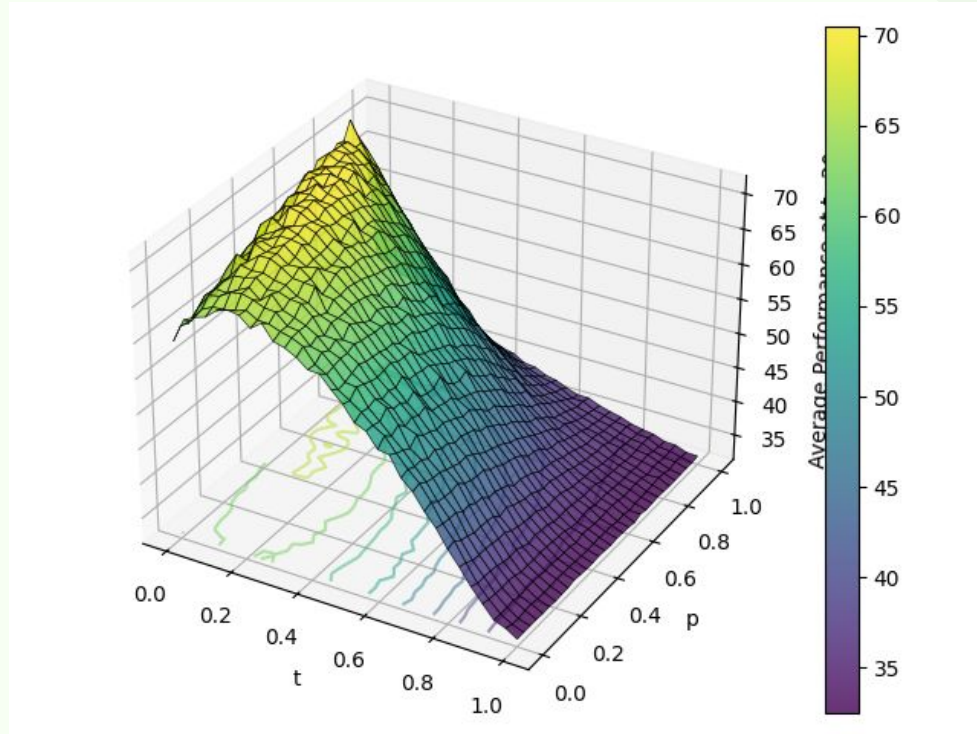
Collab-Copy Rate vs. Fitness (no turnover)



Collab-Copy Rate vs. Fitness

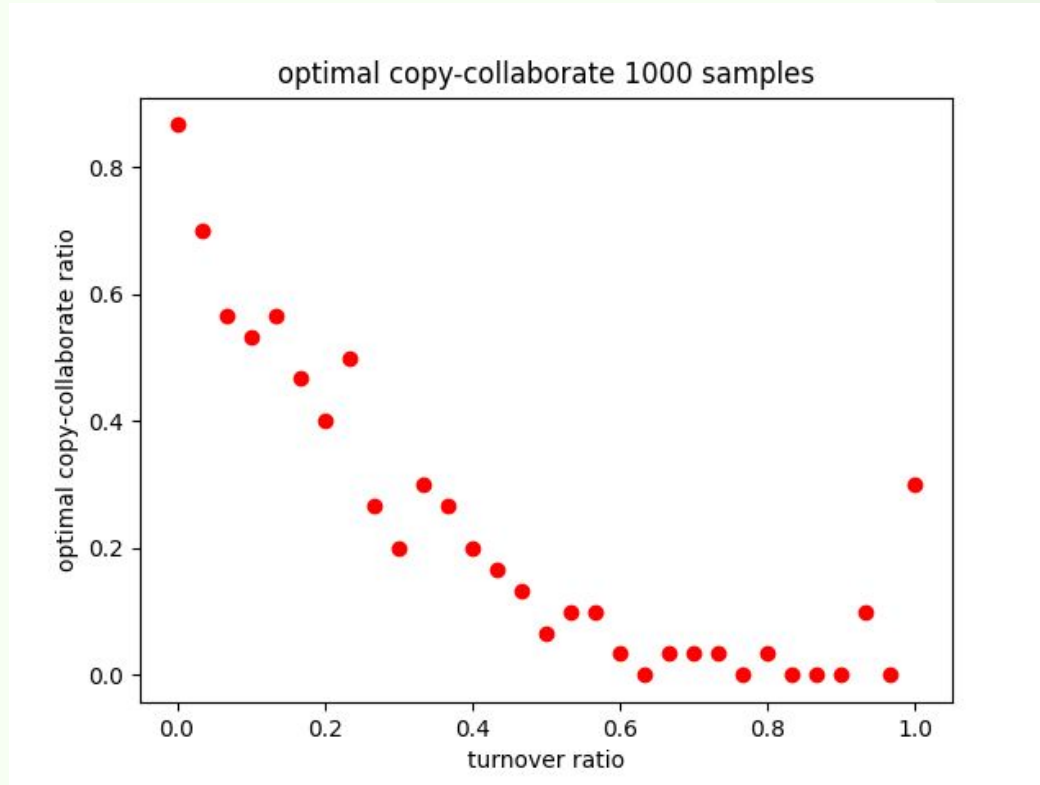


Copy-Collab Rate vs. Turnover Rate



Results

Copy-Collab Rate vs. Turnover Rate



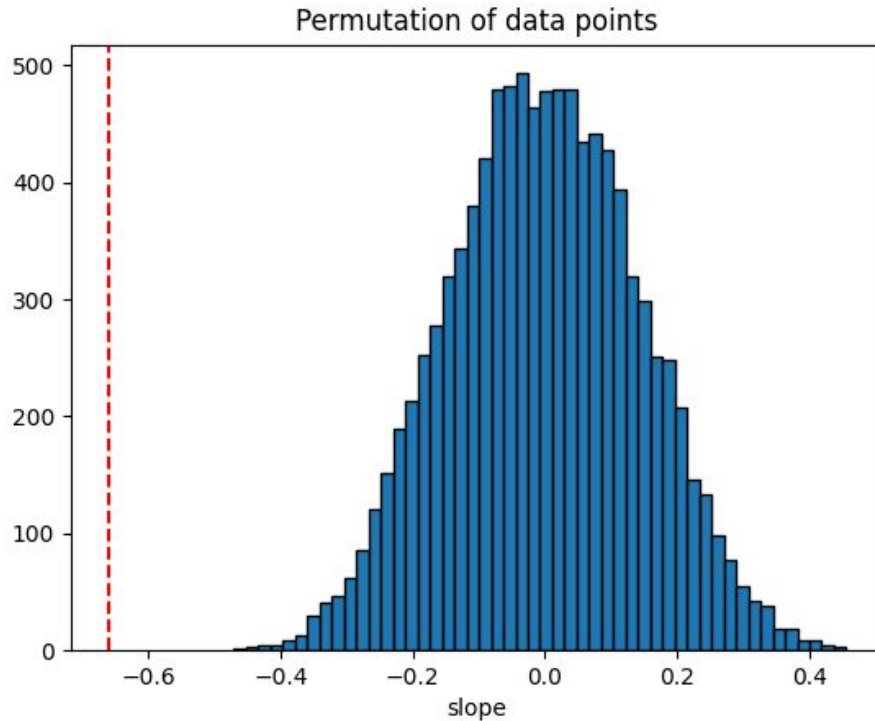
Statistics: Linear Regression

P-value: $4.2e-9$

α -value: 0.05

slope: -0.7

Permutation Histogram



Conclusion & Discussion

P-value: $4.2e-9$

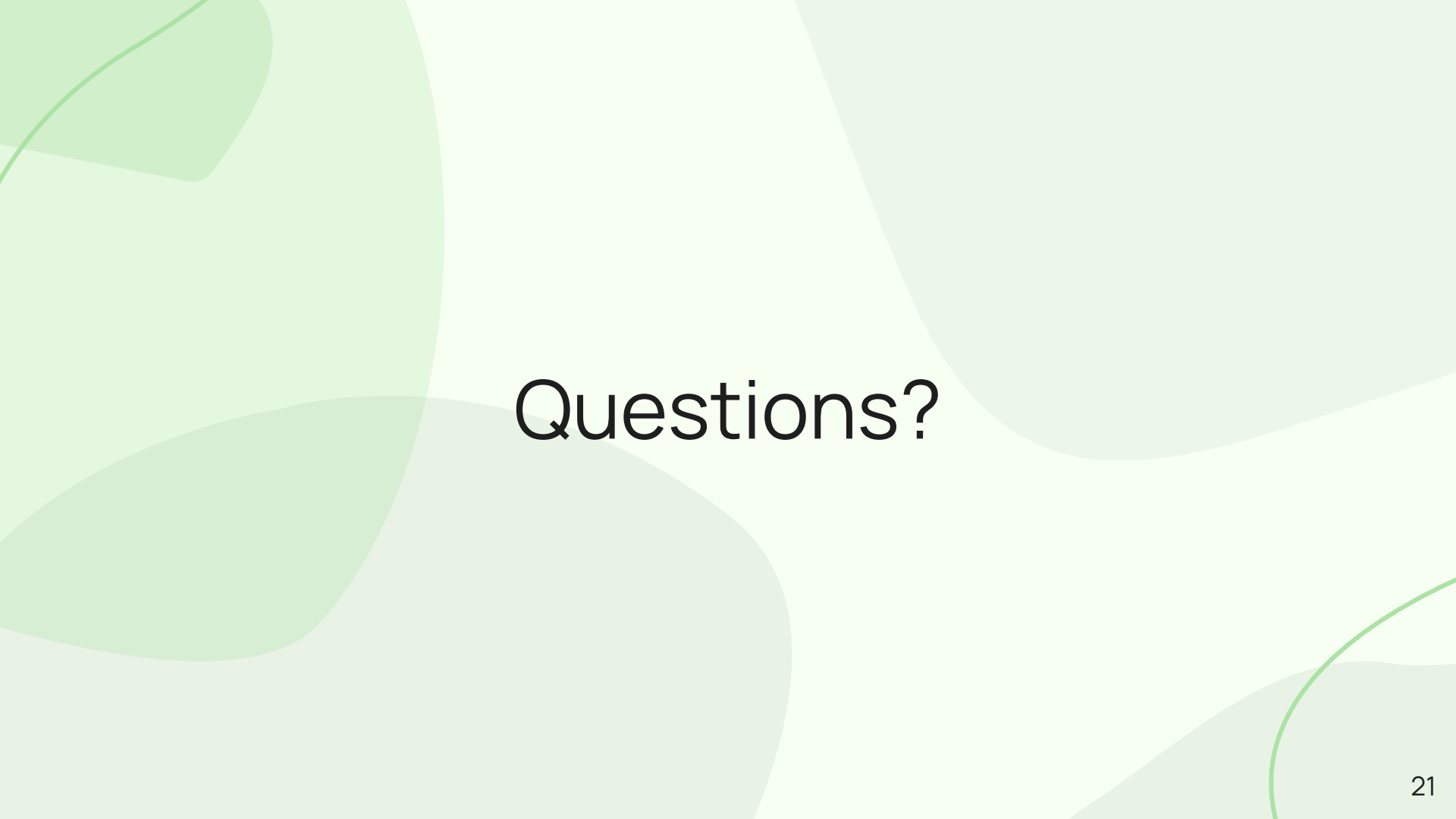
α -value: 0.05

→ reject null hypothesis

⇒ H1: Adding turnover has a statistically significant effect on the optimal collaboration-copying ratio.

Future Research

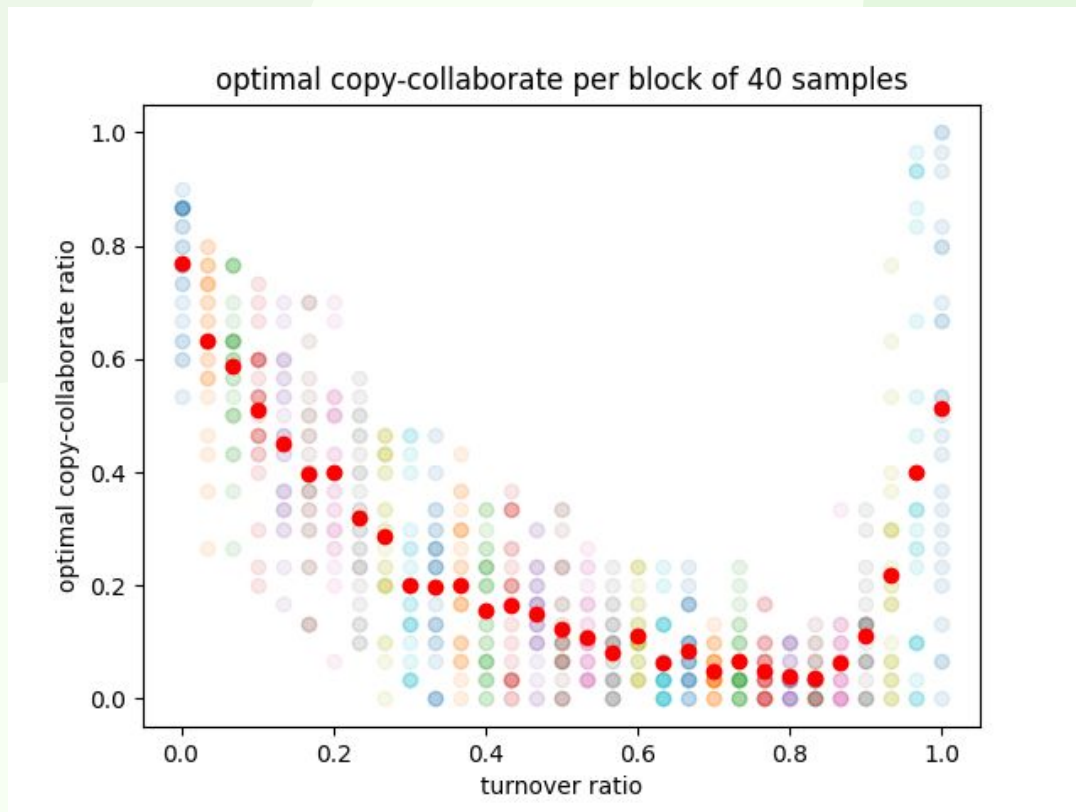
- Compare fully connected vs. partially connected networks
 - Study how sub-hubs affect escaping local optima
 - Test whether limited knowledge sharing improves global search
- Performance-based turnover (weak agents leave)
- Reinforcement learning agents
- Add more simulation steps



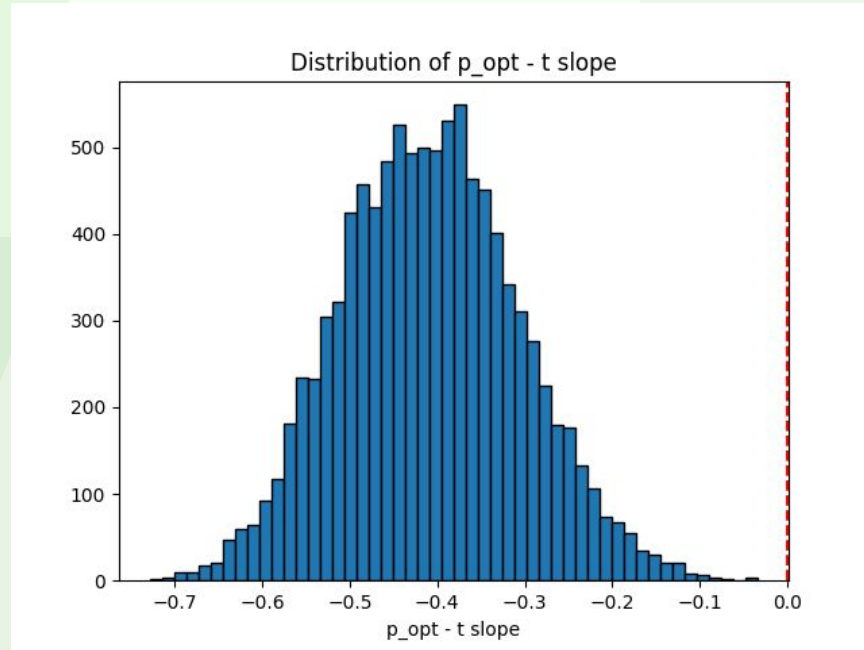
Questions?

Appendix

Copy-Collab Rate vs. Turnover Rate



Histogram



One-tailed test p-value

P-value: $2.1\text{e-}9$

Additional Slide